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PROJECT PROPOSAL

Forecast-Aware Landslide Risk Mapping for Southern Thailand

A machine learning approach using open data

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Abstract

Landslides kill thousands of people worldwide every year and recur across southern Thailand during the monsoon. The official records are incomplete, and few open tools tell authorities which slopes, and which communities, are most at risk when heavy rain is forecast. This project proposes to train a supervised classifier that ranks every 250 m cell of southern Thailand (14 provinces) by landslide susceptibility. It uses 15 free datasets, including a labelled inventory of dated landslides. A transparent rule then combines the map with the NOAA GFS rainfall forecast, judged against each tambon's own rainfall climate, and with population and building exposure, to give a live, tambon-level triage view. The plan tests seven hypotheses about preprocessing, models, factor groups, negative sampling, spatial and temporal transfer, and robustness, and analyses why approaches work or fail.

1. Problem statement

1.1 Why it matters

Landslides constitute a recurring and under-reported hazard. A global analysis of fatal non-seismic landslides identified 55,997 deaths in 4,862 events between 2004 and 2016, with Asia accounting for the largest share [1]. Over a longer period, 1998–2017, the World Health Organization reported more than 18,000 deaths and 4.8 million people affected [2]; the difference between these estimates is attributable to differences in data sources and indicates that landslide losses are incompletely recorded. In Thailand, the Department of Mineral Resources classified 54 provinces, 332 districts and 1,084 tambons as being at risk of landslides in 2019 [3]. Fatal landslides have been reported annually since 2004, although as of 2015 the EM-DAT international disaster database listed only 14 landslide events for the country, and an estimated 6,450 villages in 51 provinces were located in landslide-prone areas [4]. Southern Thailand is among the most severely affected regions. In 2011, debris flows in Krabi and Nakhon Si Thammarat resulted in 16 deaths and economic losses exceeding 10 billion baht [4], and in November 1988 a storm accompanied by widespread flooding and landslides in Nakhon Si Thammarat and Surat Thani caused approximately 350 deaths [5].

1.2 The machine learning application, users and objectives

Disaster-management authorities require information on where landslides are most likely to occur, which populations and assets are located in those areas, and when heavy rainfall is forecast. This information enables them to prioritise inspections, pre-position equipment and issue targeted warnings at the tambon level. At present, few freely available tools integrate these three elements for southern Thailand. The objectives of this project are therefore to:

1. Rank land areas by relative landslide susceptibility using open data only;
2. Quantify the population, buildings and critical facilities located on high-susceptibility ground; and
3. Identify tambons where forecast rainfall is unusually high relative to local climatic conditions.

Users: The intended users are provincial and district disaster-management offices, tambon administrations and land-use planners, and operators of road and utility networks.

1.3 Prediction Targets

The project defines two prediction targets.

Primary target (machine learning): The primary task is binary classification, in which each location is labelled either as a landslide (1) or as a sampled non-landslide point (0). The trained model will be applied to every 250 m grid cell in the study area, and the resulting scores will be grouped into five susceptibility classes, from very low to very high.

Derived target (rule-based): A live impact class will be assigned to each tambon by combining two components. The first is a terrain tier, based on the proportion of the tambon's grid cells classified as high susceptibility. The second is a local rainfall class, obtained by comparing the 24-hour GFS rainfall forecast with the tambon's wet-day rainfall percentiles from CHIRPS. This derived target is produced by a transparent rule, not by a trained model.

1.4 How success will be measured

Targets are provisional and will be fixed once the baseline is known.

Table 1. Success measures and provisional targets

Dimension	Measure	Provisional target
Ranking skill	ROC-AUC and PR-AUC under spatial block cross-validation	Best model beats logistic regression by at least 0.02 ROC-AUC
Decision value	Share of held-out landslides inside the highest-ranked area	At least 70% of held-out landslides in the top 20% of mapped area
Score quality	Brier score and calibration curve	Brier below the baseline; calibration slope near 1
Physical sense	Feature importance and partial dependence	Top drivers consistent with slope, wetness and land-use mechanisms

The targets in Table 1 are set relative to two reference points rather than chosen as absolute values. First, any model more complex than logistic regression will be considered an improvement only if it outperforms the baseline consistently across all spatial folds, by more than the variation observed between folds. The margin of 0.02 ROC-AUC is therefore provisional; it will be replaced by the observed fold-to-fold spread once the baseline has been evaluated. Second, the decision-value target is defined against a map with no skill. Such a map would, on average, contain the same share of held-out landslides as the share of area it labels high risk, that is, 20% of landslides within the top 20% of the area. Requiring at least 70% of held-out landslides within the top 20% of the area therefore means the model must concentrate observed landslides about 3.5 times more effectively than chance. Together, these two references make sure that any reported gain reflects genuine predictive skill rather than random variation or the added complexity of the model alone. Calibration is assessed against the 1:1 sampling design only. It keeps scores stable and comparable across folds, but the calibrated values are not true probabilities of landslide occurrence.

2. Proposed Solution

2.1 Data

All fifteen datasets used in this project are freely available. Most are accessed through the Google Earth Engine API, and the remainder are obtained by direct download from the data providers. The

landslide inventory is the only dataset that contains labelled observations; the others provide the explanatory variables and the contextual layers.

Table 2. Selected datasets, grouped by role [6]

Group	Datasets (source)	Role
Label	Landslide inventory (ADPC): dated landslide points, 2000–2025, plus undated records	Positive class; exclusion zone for sampled negatives
Terrain, water	ALOS AW3D30 and SRTM DEMs; HydroSHEDS 15ACC and MERIT Hydro	Slope, curvature, TRI, TPI, LS factor, upstream area, TWI, SPI
Ground, surface	OpenLandMap soil; NDVI from Landsat 5/7/8/9 and Sentinel-2; SERVIR-Mekong land cover; GLiM lithology; OSM roads and rivers; GEM active faults	Soil strength, vegetation, land use, rock type, disturbance
Rainfall	GPM IMERG (antecedent rain), NOAA GFS (forecast), CHIRPS daily (1991–2020 climate)	Trigger experiment; live forecast; local percentiles
Exposure, units	WorldPop 2020 constrained; OSM buildings and facilities; GADM 4.1 boundaries	People and assets at risk; tambon join key; CV groups

Held in reserve: Sentinel-1 radar, ERA5-Land soil moisture, Hansen forest loss, Dynamic World, OpenLandMap soil organic carbon.

2.2 Data integration and preprocessing

Data integration is performed at three spatial levels. At the point level, all datasets are linked to the landslide and non-landslide locations using geographic coordinates and the corresponding event date, forming the training dataset. At the grid level, the same variables are extracted for each cell of a regular 250 m grid covering the study area, providing the basis for the susceptibility map. At the administrative level, outputs are aggregated by tambon using the GADM identifier to support the impact assessment. To prevent temporal information leakage, only data recorded prior to each event are used in deriving its explanatory variables; accordingly, time-dependent variable such as NDVI and land cover are obtained from observations preceding the event date (for NDVI, a composite from the year before; see 2.2.3). The data preparation procedure follows the six steps in Figure 1; encoding and feature selection are described separately in 2.2.7 and 2.2.8.

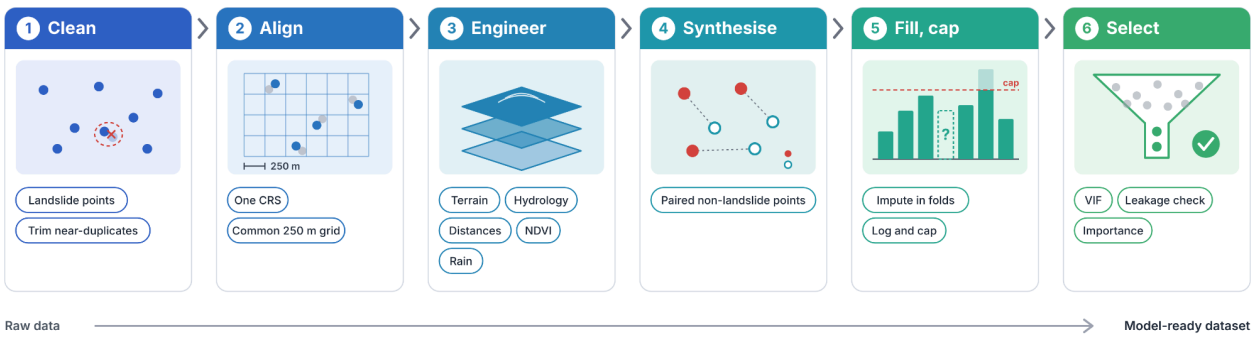


Figure 1. Data preparation pipeline

2.2.1 Data cleaning and deduplication

Mapped landslide inventories commonly contain near-duplicate records located only a few metres apart, occasionally with different dates. If retained, one record of such a pair could be assigned to the training set and the other to the test set, artificially inflating model performance. To prevent this, spatial thinning will be applied so that only one point is retained per cluster. A further check will confirm that no test point lies closer to a training point than the validation buffer distance.

2.2.2 Spatial alignment and resolution

The input datasets differ substantially in spatial resolution: approximately 30 m for terrain, 90–460 m for hydrology, 250 m for soil, 0.05° for CHIRPS rainfall and 0.25° for GFS rainfall. All layers will be projected to a common coordinate reference system, and features will be extracted at identical point locations and on a single 250 m grid, so that every record represents the same spatial footprint. Coarse-resolution layers will be treated as contextual variables rather than as indicators of fine-scale variation.

2.2.3 Feature engineering

Conditioning factors will be derived from several sources. Elevation, slope, aspect, general and profile curvature, the terrain ruggedness index (TRI), the topographic position index (TPI) and the slope length–steepness (LS) factor will be computed from the digital elevation model. Upstream contributing area, the topographic wetness index (TWI) and the stream power index (SPI) will be derived from the hydrological datasets. Distances to the nearest road, river and active fault will also be calculated. Vegetation conditions will be represented by a median NDVI composite from the year before each recorded date, rather than from the weeks immediately before it. Because the recorded dates are likely image dates rather than event dates (Section 3.1), a short pre-event window could include the landslide scar itself. For the susceptibility map, the same composite will be computed from the most recent full year.

2.2.4 Synthetic negative samples

Because the landslide inventory records only the presence of landslides, the negative class must be generated. For each landslide, one non-landslide point will be sampled 2 to 10 km away in the same province, kept clear of all recorded landslides and of other negative points, and assigned the date of its paired landslide. This paired design keeps both classes within comparable regions and seasons, reducing the risk that the model learns a simple contrast between mountainous and flat terrain. Some contrast may nevertheless remain, because points sampled near a landslide tend to fall on lower or gentler ground; experiment E4 will quantify this effect by comparing the paired design with randomly placed and environmentally matched negatives. As absence is assumed rather than observed, model outputs will be interpreted as a relative ranking of susceptibility, not as a probability of occurrence. Because each negative shares the date and rainfall history of its paired landslide, the design is suitable for the susceptibility model but not for a rainfall-trigger model, which would require negatives drawn from the same locations on dates without landslides.

2.2.5 Missing values

Missing values are expected in NDVI under cloud cover and in soil variables near water bodies. They will be imputed using the median or k-nearest neighbours, with the imputation fitted within each training fold and a binary missing-value indicator added. This prevents information from the test fold from influencing the training process.

2.2.6 Outlier treatment

Upstream area, SPI and TWI are expected to be strongly right-skewed. These variables will be log-transformed and their extreme tails capped. Physically plausible extremes, such as very steep slopes, will be retained because they carry genuine information about landslide susceptibility.

2.2.7 Encoding and scaling

Land cover and lithology are categorical variables; one-hot encoding and Weight-of-Evidence encoding will be compared in experiment E1. Feature scaling will be applied only to models that are sensitive to feature magnitude, namely logistic regression, support vector machines and k-nearest neighbours, and will be fitted within each training fold to avoid information leakage. Tree-based models will be trained on unscaled features.

2.2.8 Feature selection

Redundant terrain indices will be removed through correlation analysis and variance inflation factor (VIF) screening. A leakage check will exclude any variable derived from information recorded after the event date. NDVI windows close to the recorded date are excluded for the same reason, since they may capture vegetation loss caused by the landslide. The final feature set will be determined by permutation importance and by the factor-group ablation in experiment E3. Variables that may primarily reflect mapping bias rather than physical susceptibility, such as distance to roads, will be tested explicitly.

2.3 Model selection

The task is tabular binary classification on a few thousand labelled points per class, with mixed numeric and categorical inputs. That suits a transparent baseline and tree ensembles better than deep learning, which needs far more data and is hard to explain to officials.

Table 3. Candidate models and why each is appropriate

Tier	Models	Why appropriate
Baseline	Logistic regression; frequency ratio / Weight-of-Evidence	Interpretable, the standard reference in landslide studies; sets the bar every other model must beat
Simple checks	k-NN, decision tree, naive Bayes	Cheap checks of how much complexity actually helps
Ensembles	Random forest, XGBoost, LightGBM, AdaBoost	Capture non-linear effects and interactions such as slope × wetness; strong on tabular data

Tier	Models	Why appropriate
Kernel	SVM with RBF kernel	Smooth non-linear boundary in a compact, dense feature space

All models share one scikit-learn pipeline (imputation, encoding, scaling, model), so preprocessing is refitted inside every fold. Scores of the selected model will be calibrated (isotonic or Platt) so that the relative classes mean the same thing across the map.

2.4 Experimental design

Each experiment (in Table 4) is written as a hypothesis with an expected outcome, so a result confirms or refutes something about the problem instead of only ranking models. All comparisons use identical spatial folds; hyperparameters are tuned by random search inside inner folds (nested cross-validation).

Table 4. Planned experiments, hypotheses and designs

Experiment	Hypothesis (expected outcome)	Design
E1 Preprocessing	Imputation and encoding choices change rankings little	Median vs k-NN imputation; one-hot vs WoE; with and without log transforms
E2 Model family	Boosting beats logistic regression, but only by a small margin	All models on the same folds with nested tuning
E3 Ablation	Terrain carries most of the signal; NDVI, land cover and roads add some	Terrain only, then add hydrology, soil, vegetation, land cover, lithology, distances
E4 Negative sampling	Random negatives inflate scores relative to paired ones	Paired ring vs random vs environment-matched negatives
E5 Spatial transfer	Scores drop in provinces the model has never seen	Leave-one-province-out
E6 Temporal check	A model of earlier events still ranks later events	Train on earlier event years, test on later years (cut-off set after the data audit)
E7 Robustness	5% label noise and a DEM swap move rankings little	Flip 5% of labels; ALOS vs SRTM terrain

Validation: Nearby points are alike, so random splits overstate accuracy. We will use spatial block cross-validation (0.5° blocks) with a buffer around each test block, plus a held-out region reported once at the end.

Metrics: ROC-AUC and PR-AUC for ranking; sensitivity, specificity and balanced accuracy at a chosen threshold; Brier score and reliability curves for calibration; and the share of held-out landslides in the top 10% and 20% of area for decision value. We report the mean and spread across folds, not a single number.

2.5 Analysis

The analysis aims to explain why particular approaches succeed or fail, and to identify the conditions under which the susceptibility map should not be relied upon.

- **Error analysis:** False negatives and false positives will be mapped, and their distribution will be tested for clustering by province, lithology, slope class, land cover and distance to roads. Systematic clustering would indicate where the model, or the underlying data, is deficient.

- **Drivers and interpretability:** The contribution of each conditioning factor will be assessed using permutation importance, SHAP values and partial-dependence curves. These results will be checked against established geotechnical understanding, in particular the expected effects of slope, soil wetness and land use. The interaction between slope and the topographic wetness index will also be examined to test whether steep slopes are hazardous mainly when they are also wet.

- **Performance across groups:** Model performance will be reported separately by province, land-cover class, lithology and elevation band. Large differences between groups would show where the model generalises poorly, and experiment E5 will indicate how well it transfers to provinces it has not seen.

- **Sensitivity:** The robustness of the conclusions will be tested by varying the negative-sampling distance, grid resolution, source digital elevation model and level of label noise, and checking whether the main findings remain unchanged.

- **Failure conditions:** Situations in which the model is likely to be unreliable will be identified explicitly. These include provinces without labelled events, records affected by cloud-related gaps, steep road cuts and areas represented by coarse soil data.

- **Uncertainty:** The variation in predicted scores across cross-validation folds will be mapped, so that users can see where the susceptibility ranking is uncertain.

3. Additional considerations

3.1 Data quality and limitations

- **Representativeness:** A preliminary review of the inventory shows that dated landslide records are available for 8 of the 14 provinces, and that 99.9% of these records are concentrated in six of them (Figure 2). The other six provinces (Phuket, Songkhla, Satun, Pattani, Yala and Narathiwat) contain no labelled events, so predictions for these areas will be extrapolations beyond the training data. Experiment E5 addresses this limitation by holding out each labelled province in turn to estimate how much predictive performance declines in areas the model has not seen. Areas without training labels will also be clearly identified on the dashboard so that users can interpret those predictions with appropriate caution.

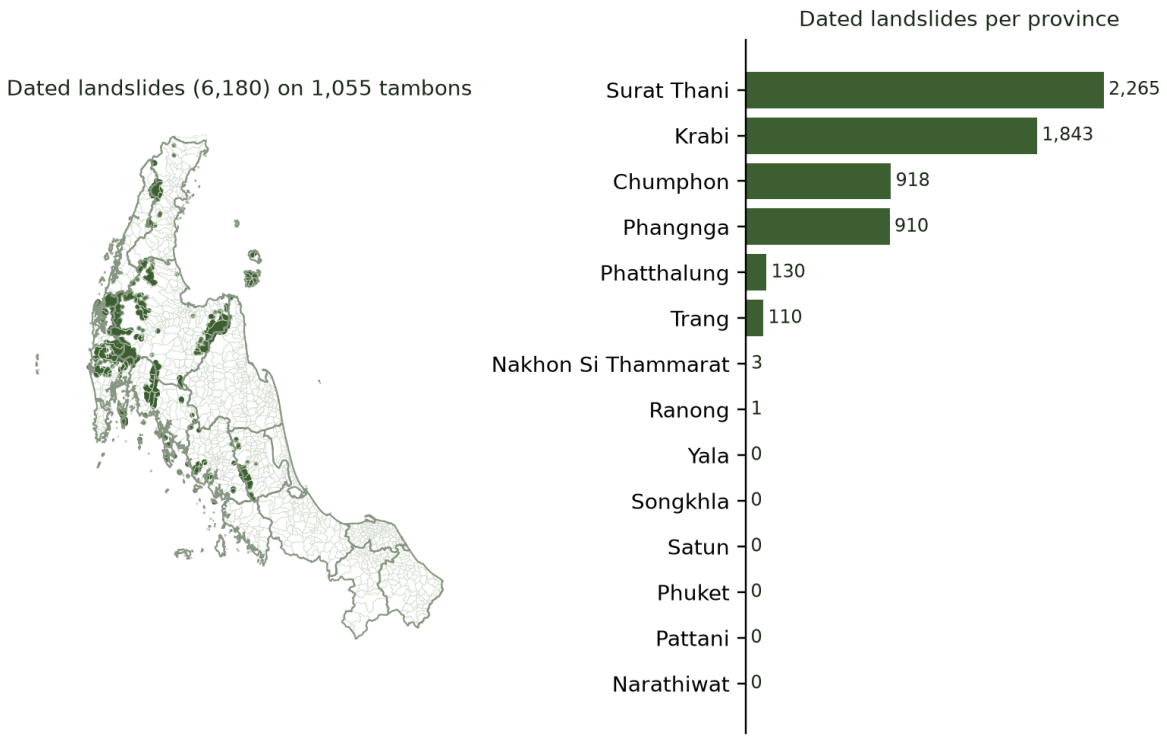


Figure 2. Dated landslides in the study area, by province (initial data audit)

- **Label quality:** The dates recorded in the inventory appear to correspond to the acquisition dates of the satellite images used for mapping rather than to the actual dates of landslide occurrence. This limits the reliability of any model that links landslides to specific rainfall events. Susceptibility modelling does not depend on precise event timing and is therefore retained as the primary target.
- **Sufficiency:** Several thousand labelled points described by approximately 20 conditioning factors provide sufficient data for the tabular models proposed. The main constraint on model reliability is the limited spatial coverage of the labels rather than the number of records.
- **Resolution:** The GFS forecast (about 28 km) and the soil layers (250 m) are much coarser than the terrain data (30 m). The live impact view should therefore be treated as a screening tool rather than a site-level warning.

3.2 Practical value

The map tells officials where to look first; the live view adds when, by flagging tambons where unusual rain meets susceptible, populated ground. A what-if slider lets planners test a storm (for example 150 mm in 24 h). The impact rule (Table 5) is deliberately transparent:

Table 5. Impact rule: terrain tier combined with forecast rain against the local normal

Terrain tier \ rain vs local normal	< P75	P75–P90	P90–P95	> P95
Low	Low	Low	Moderate	Moderate
Moderate	Low	Moderate	High	High
High	Low	Moderate	High	Very high

It is a triage rule tested for consistency, not a validated forecast; local officials make the final decision.

3.3 Deployment plan (AWS) and CI/CD

The system will be deployed on Amazon Web Services (AWS) using a containerised architecture. A FastAPI backend will expose endpoints for model predictions, the susceptibility map and the rainfall scenario tool, while a Next.js web dashboard will present the map and an early-warning page to users. Both components will run as Docker containers on an AWS EC2 instance, with nginx providing HTTPS access and Traefik handling request routing between services.

Continuous integration and deployment will be implemented with GitHub Actions. Each change pushed to the repository will trigger an automated pipeline that runs the test suite, builds the container images and deploys them to the server over SSH. A post-deployment health check will verify that the service is running correctly, and the previous version will be restored automatically if the check fails.

The model will follow a separate pipeline (Figure 3). Training data will be versioned, each trained model will be stored together with its metadata, and a model will be promoted to production only if it outperforms the current version under spatial cross-validation.

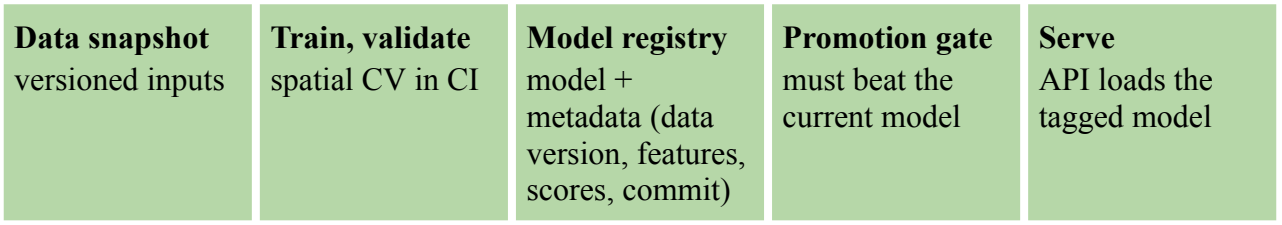


Figure 3. Model CI/CD pipeline

Model management will follow the same version-control principles as the application code. Each trained model will be stored together with its metadata, including the version of the input data, the feature set, the validation scores and the corresponding code revision. A model will be promoted to production only if it passes a predefined performance gate. Retraining will be triggered by any of three conditions: the addition of a new batch of inventory records, a scheduled quarterly review, or a detected shift in the distribution of input variables. In addition, a scheduled job will update the GFS rainfall forecast and the derived impact layer at regular intervals.

3.4 Risks and alternatives

Table 6. Main risks and backup plans

Risk	Effect	Backup plan
Inventory dates reflect satellite-image acquisition rather than event occurrence	The timing of rainfall triggers cannot be reliably established	Susceptibility will be retained as the primary target; rainfall-trigger modelling will be treated as an optional experiment because it requires separate negative samples from the same locations on non-landslide dates
Non-landslide points are assumed rather than observed	Scores indicate relative ranking, not true probabilities of occurrence	Alternative sampling designs will be compared (E4), and results will be reported as rankings and calibrated classes

Risk	Effect	Backup plan
Mapping bias towards accessible areas near roads	The model may learn landslide visibility rather than physical instability	Road distance will be removed in an ablation test (E3), and predicted patterns will be inspected for alignment with roads
Missing labels in six provinces	Model performance in these areas cannot be measured directly	Transfer to unseen areas will be estimated by leave-one-province-out validation (E5), and extrapolated areas will be flagged on the dashboard
Google Earth Engine usage quotas and restricted asset access	Data extraction may be delayed or interrupted	Requests will be batched and made resumable; Dynamic World will serve as a substitute land-cover source

3.5 Project management

The project will be carried out over eight weeks by a team of five members, with responsibilities divided as follows (roles to be confirmed): M1, landslide inventory and negative sampling; M2, extraction of conditioning factors and preprocessing; M3, rainfall data, exposure layers and the impact rule; M4, modelling and experiments; and M5, analysis, deployment and the CI/CD pipeline. Work will be coordinated through GitHub, using issues to track tasks, pull-request reviews for all changes and a protected main branch. The team will hold a weekly check-in to review progress against milestones, and will share documents and data through a common drive. The timeline and milestones are shown in Table 7.

Table 7. Eight-week timeline, member and milestones

Workstream	Member	W1	W2	W3	W4	W5	W6	W7	W8
Data access and audit	M1, M2								
Preprocessing	M2, M1								
Baselines and CV set-up	M4								
Experiments E1–E7	M4, M5								
Analysis and interpretation	M5, M4								
Impact layer and app	M3, M5								
Report and slides	All								
Milestones			**Data ready**	**Baseline**		**E1–E7**	**Analysis**	**App demo**	**Report**

4. Expected contributions

The project is expected to make three contributions.

- First, it will deliver a reproducible data pipeline built entirely from open sources, integrating fifteen freely available datasets by location, date and administrative unit (tambon). Explicit rules for preventing information leakage and removing duplicate records will be documented so that the workflow can be repeated and extended.
- Second, it will provide evidence on what drives model performance rather than a single accuracy figure. Pre-stated hypotheses will be tested on the effects of preprocessing choices,

model complexity, groups of conditioning factors, negative-sampling design, and transfer across space and time. The analysis will examine why particular approaches succeed or fail.

- Third, it will produce a working prototype of a live impact view, deployed on AWS. The prototype combines the susceptibility map with forecast rainfall assessed against local climate and with population and building exposure, and includes a what-if scenario tool for testing hypothetical rainfall events.

References

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